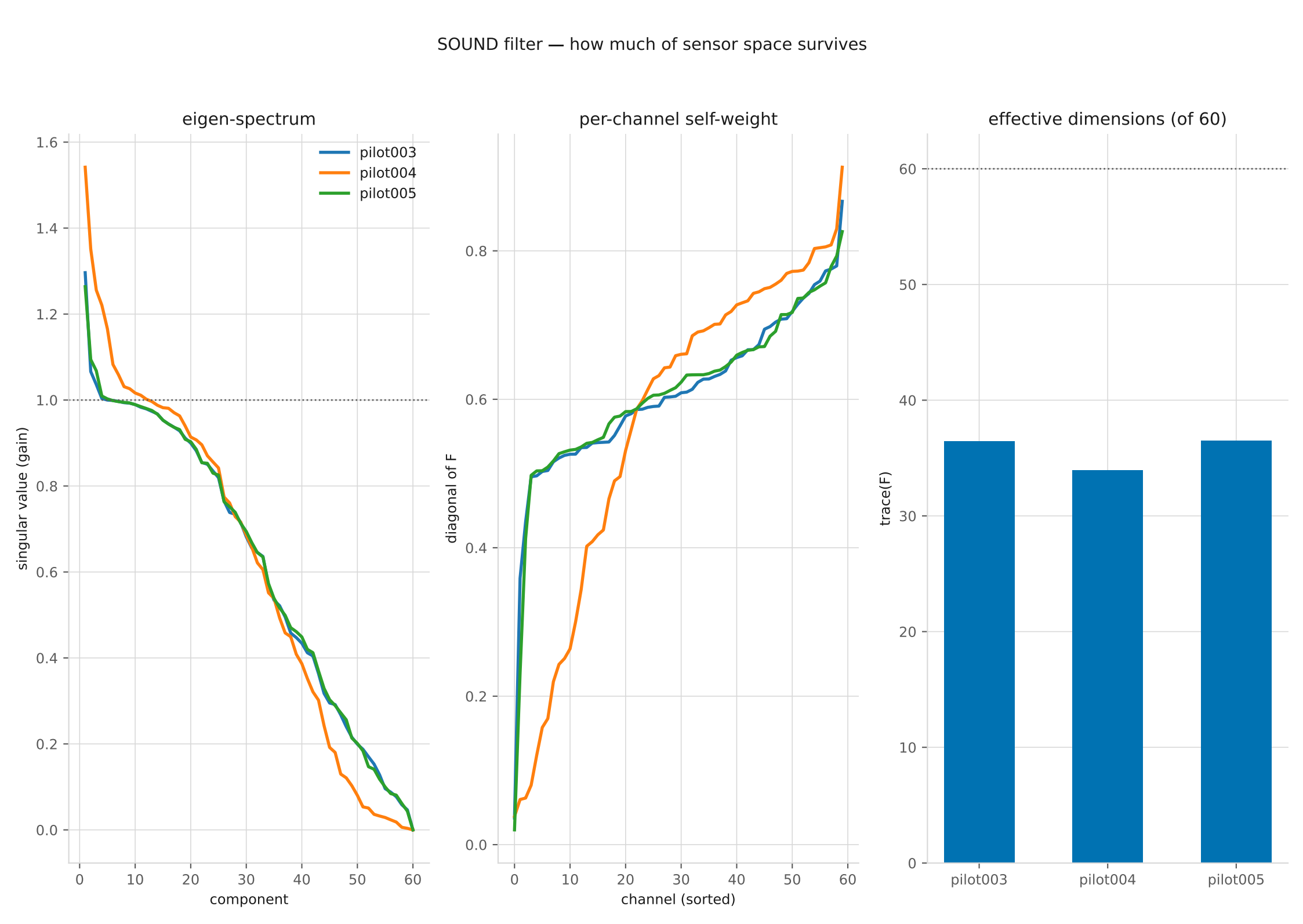
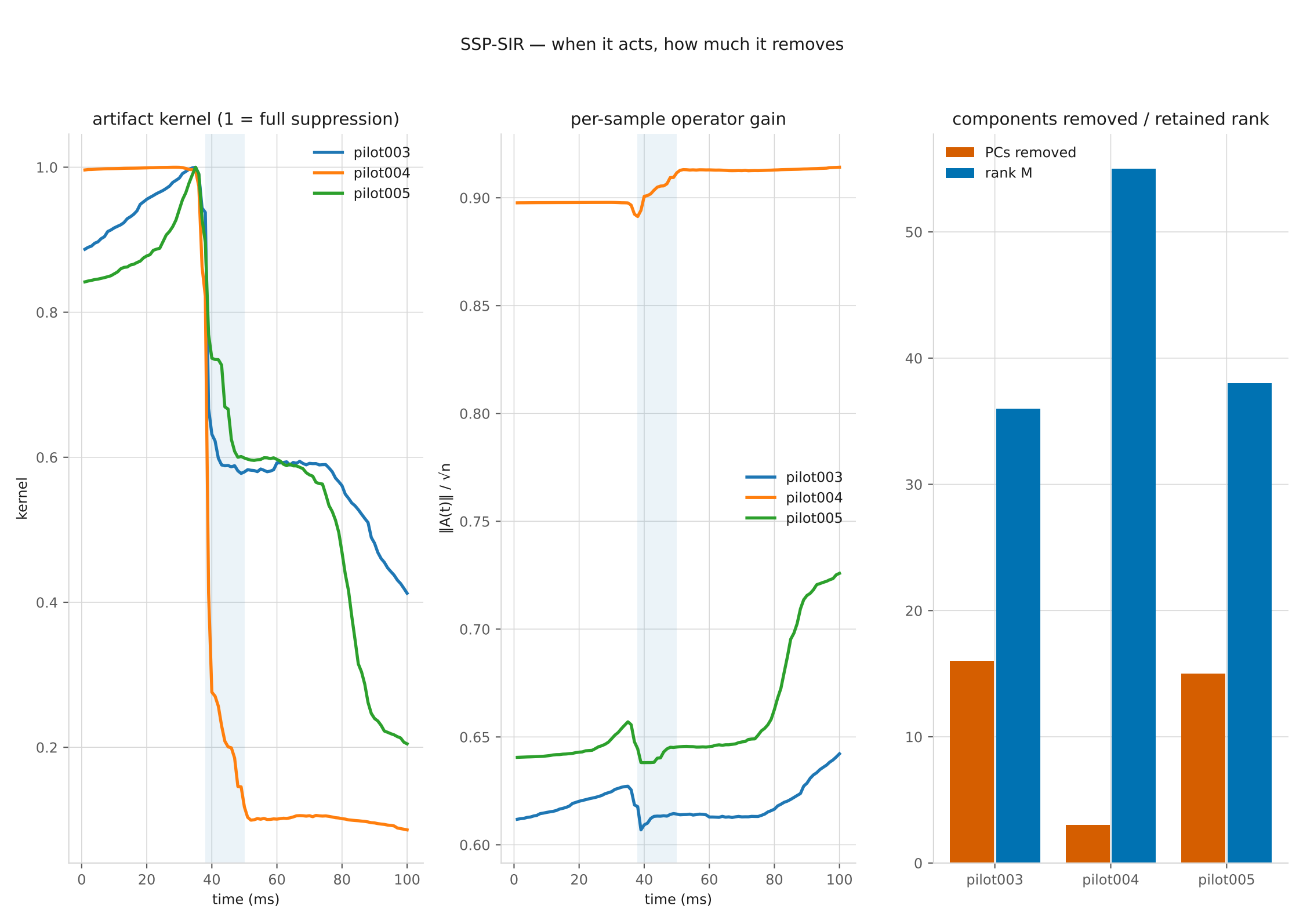
# SOUND and SSP-SIR — online post-stimulus chain, replayed

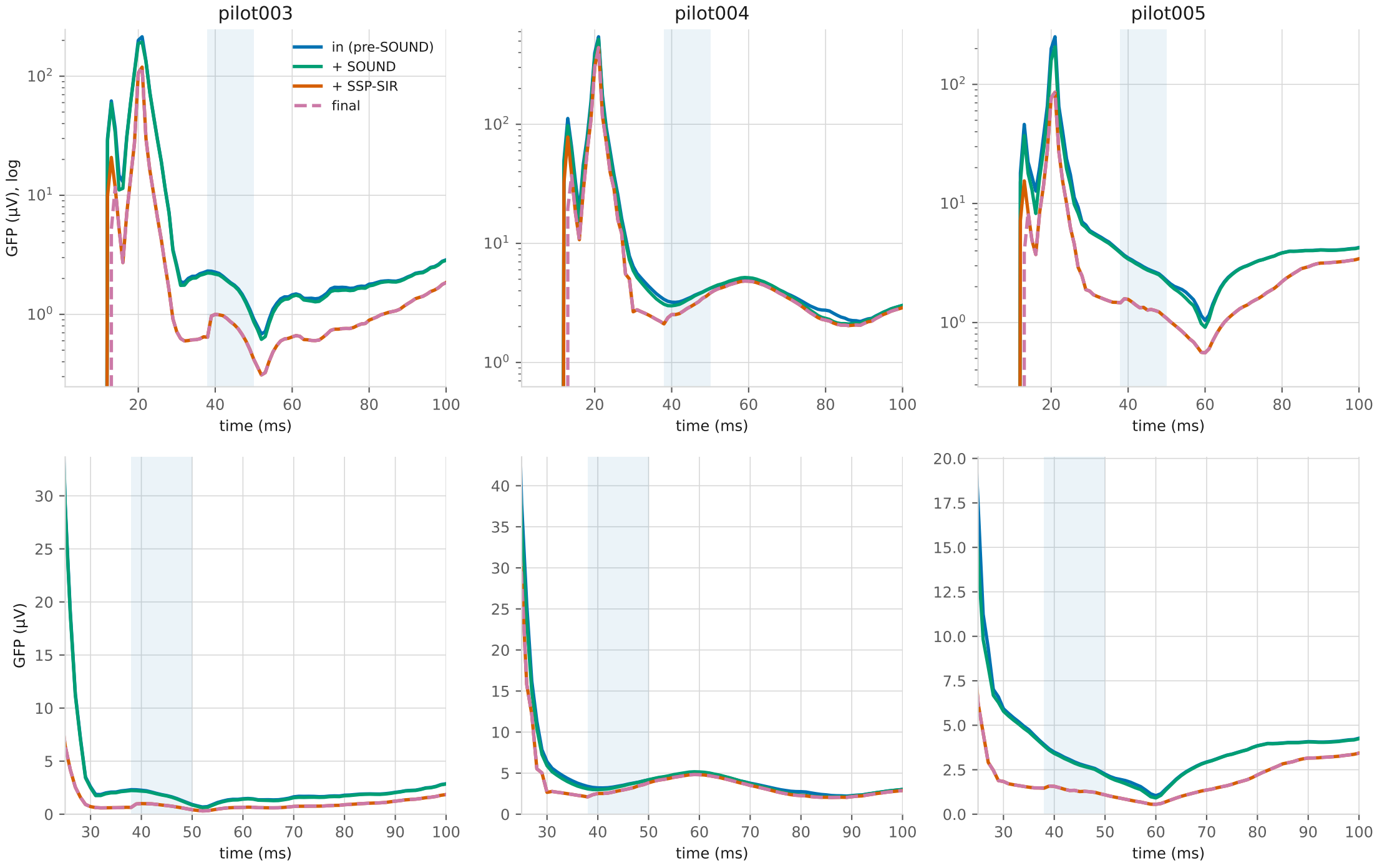
stages: pre_sound (SOUND input) -> sound -> sspsir -> final (== *_post_proc.npy)
trial groups: baseline, calibration, intervention, evaluation

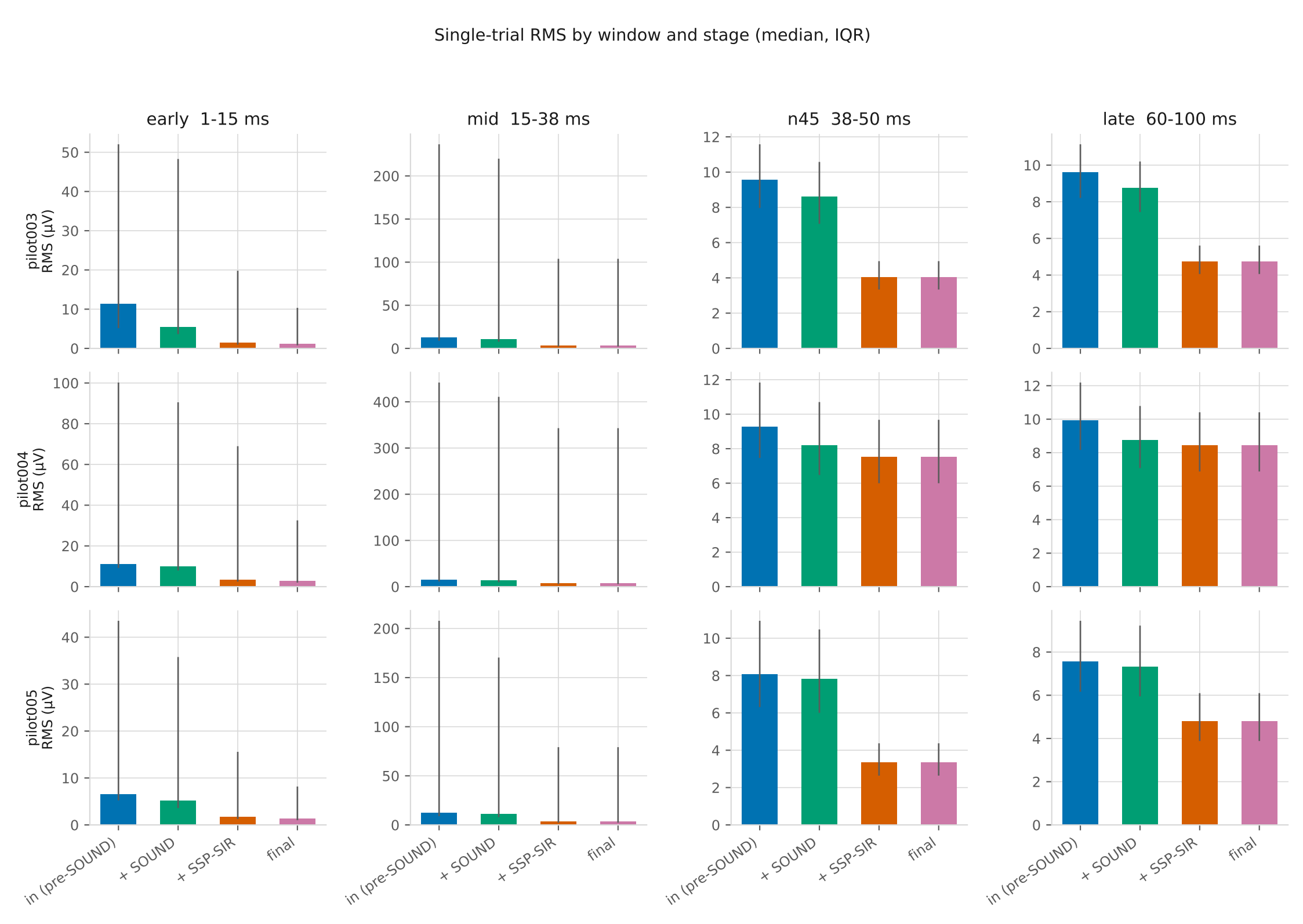
| session | trials | replication | amp check | vs online | crop win | fit win | SOUND dof | \|F-I\| | SSP PCs | rank M | kernel@fit | bad channels |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| pilot003 | 1425 | 6.48e-12 | 0.00e+00 | 5.07e-15 | 38-50 | 41-46 | 36.4 | 0.536 | 16 | 36 | 0.626 | C3,CP3,FC1,P4,T7 |
| pilot004 | 1425 | 5.40e-12 | 0.00e+00 | 5.00e-10 | 38-50 | 44-49 | 34.0 | 0.651 | 3 | 55 | 0.267 | Fz,P4,P8 |
| pilot005 | 1425 | 4.55e-12 | 0.00e+00 | 4.10e-15 | 38-50 | 39-42 | 36.5 | 0.538 | 15 | 38 | 0.690 | C3,CP3,T8 |

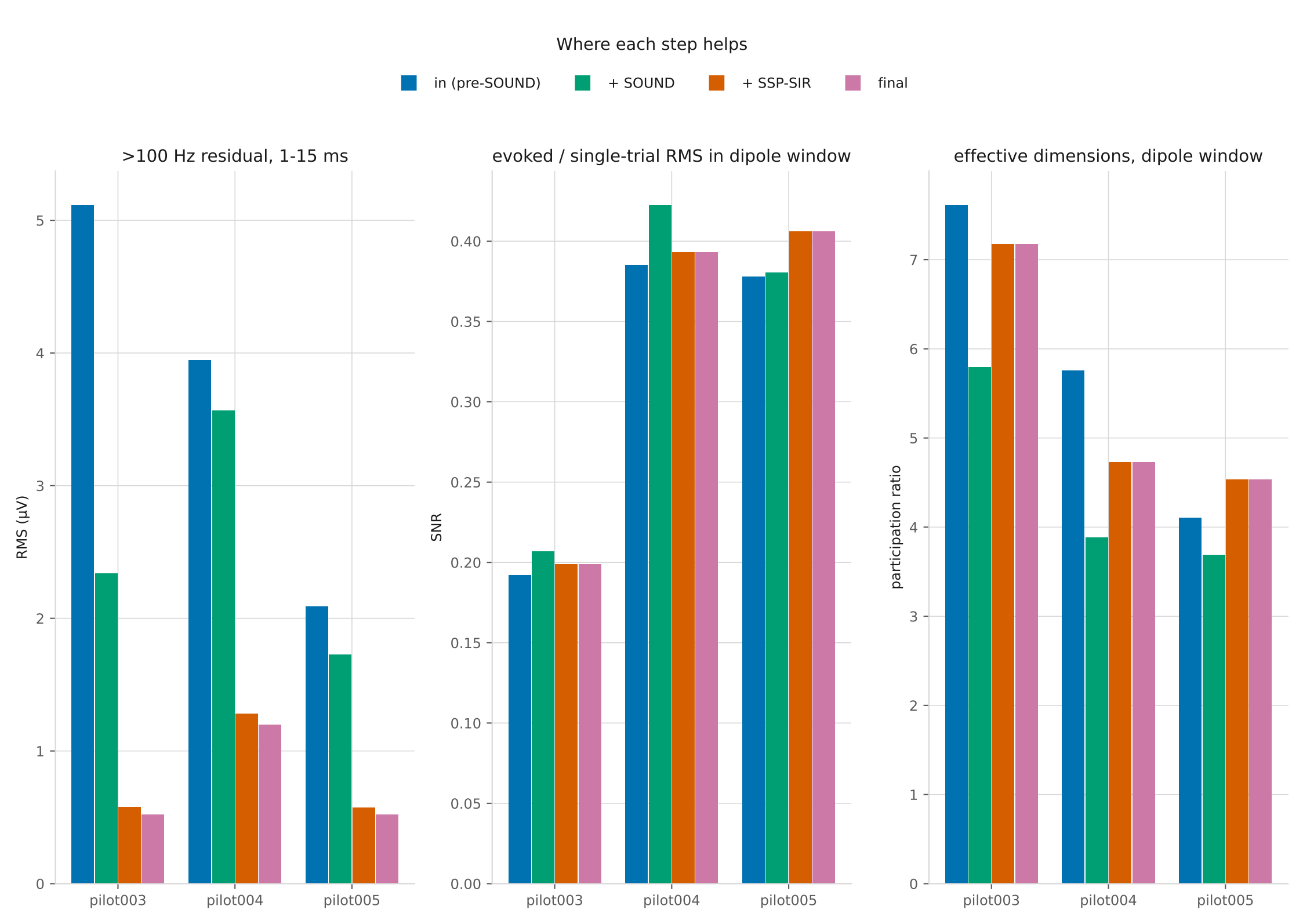
replication = max |replayed final − saved *_post_proc.npy|  (expect ~1e-12 or 0)
amp check   = max |own dipole fit − DipoleFitter.fit_trial|  (expect ~1e-15)
vs online   = max |fit_trial − tep_amplitude_raw in the trial table|
SOUND dof = trace of the filter, out of 60 channels; |F-I| = RMS distance from identity

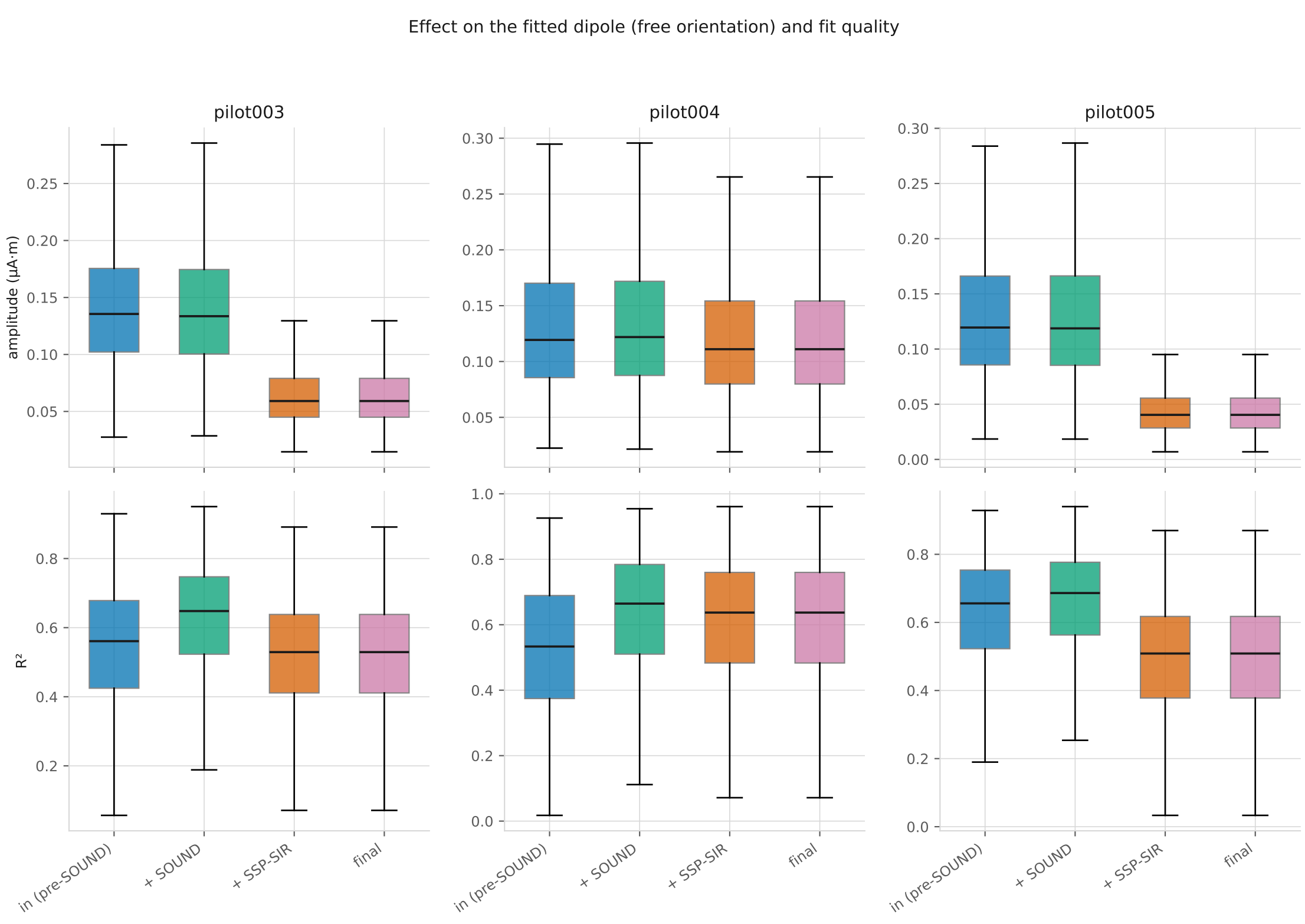

SOUND filter — how much of sensor space survives
eigen-spectrum
pilot003
pilot004
pilot005
singular value (gain)
component
per-channel self-weight
diagonal of F
channel (sorted)
effective dimensions (of 60)
trace(F)
pilot003
pilot004
pilot005


SSP-SIR — when it acts, how much it removes
artifact kernel (1 = full suppression)
pilot003
pilot004
pilot005
kernel
time (ms)
per-sample operator gain
‖A(t)‖ / √n
time (ms)
components removed / retained rank
PCs removed
rank M
pilot003
pilot004
pilot005

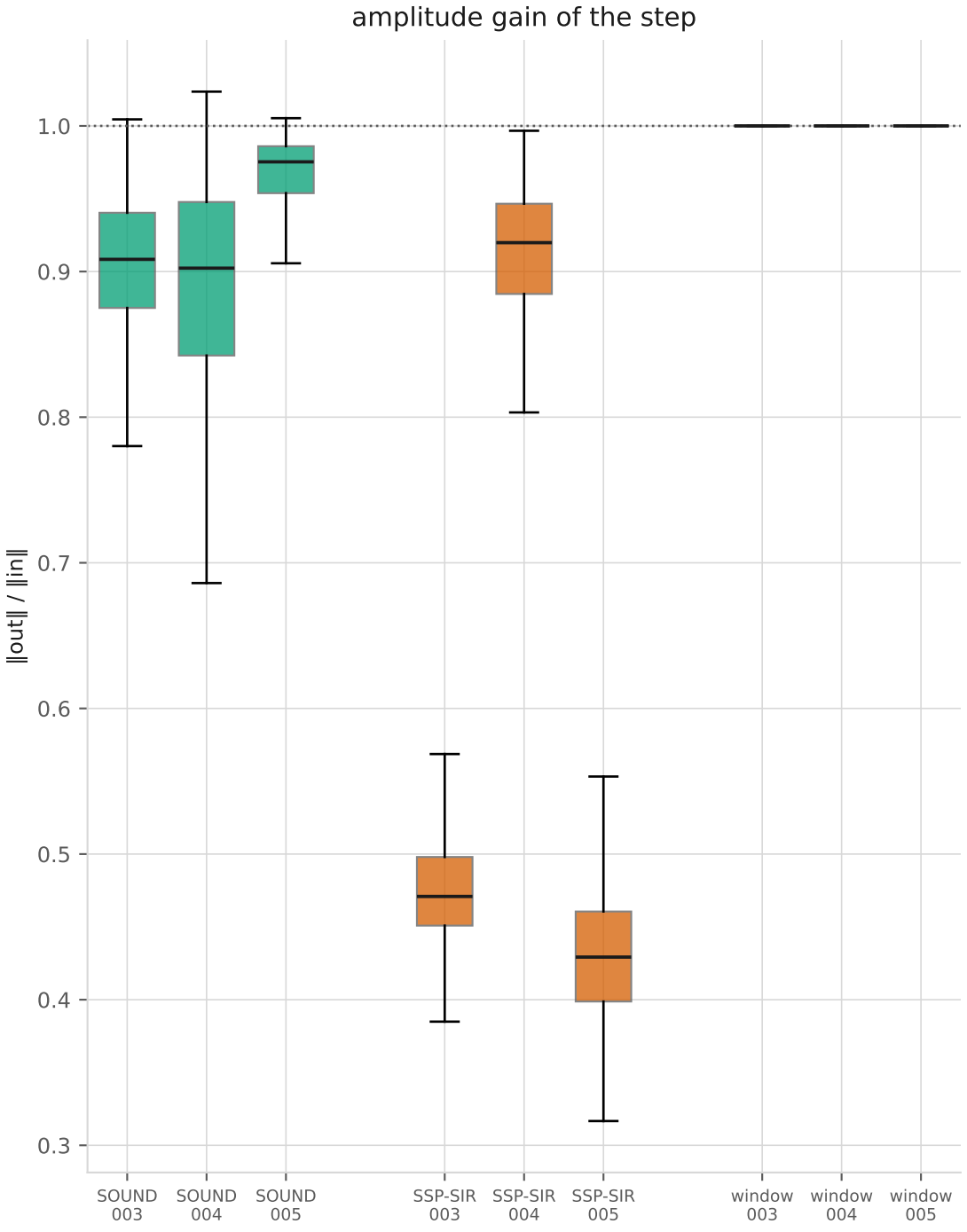

Evoked GFP by stage — whole window (log) and the TEP range
pilot003
pilot004
pilot005
in (pre-SOUND)
+ SOUND
+ SSP-SIR
final
GFP (µV), log
GFP (µV)
time (ms)

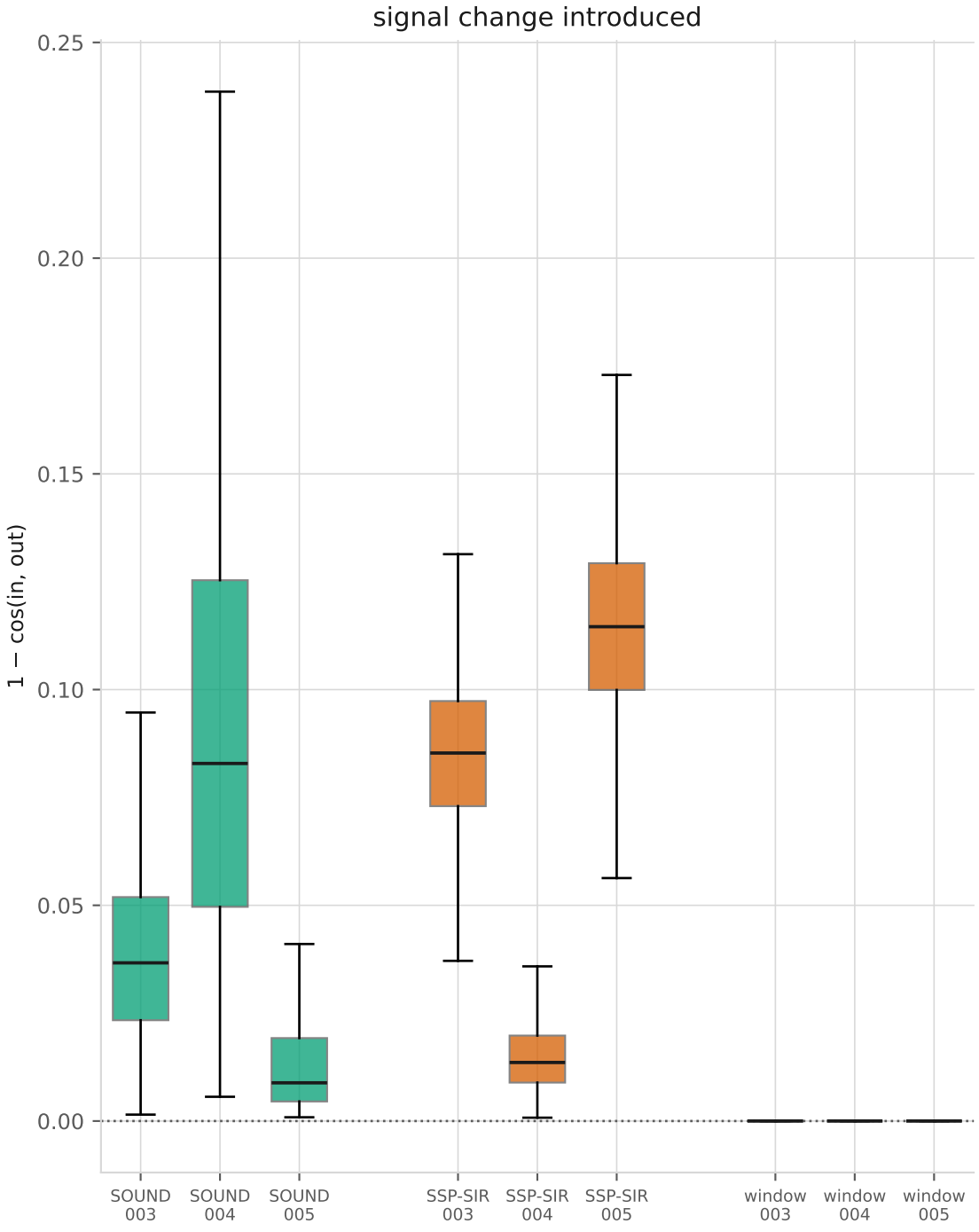

Single-trial RMS by window and stage (median, IQR)
early 1-15 ms
mid 15-38 ms
n45 38-50 ms
late 60-100 ms
pilot003 RMS (µV)
pilot004 RMS (µV)
pilot005 RMS (µV)
in (pre-SOUND)
+ SOUND
+ SSP-SIR
final

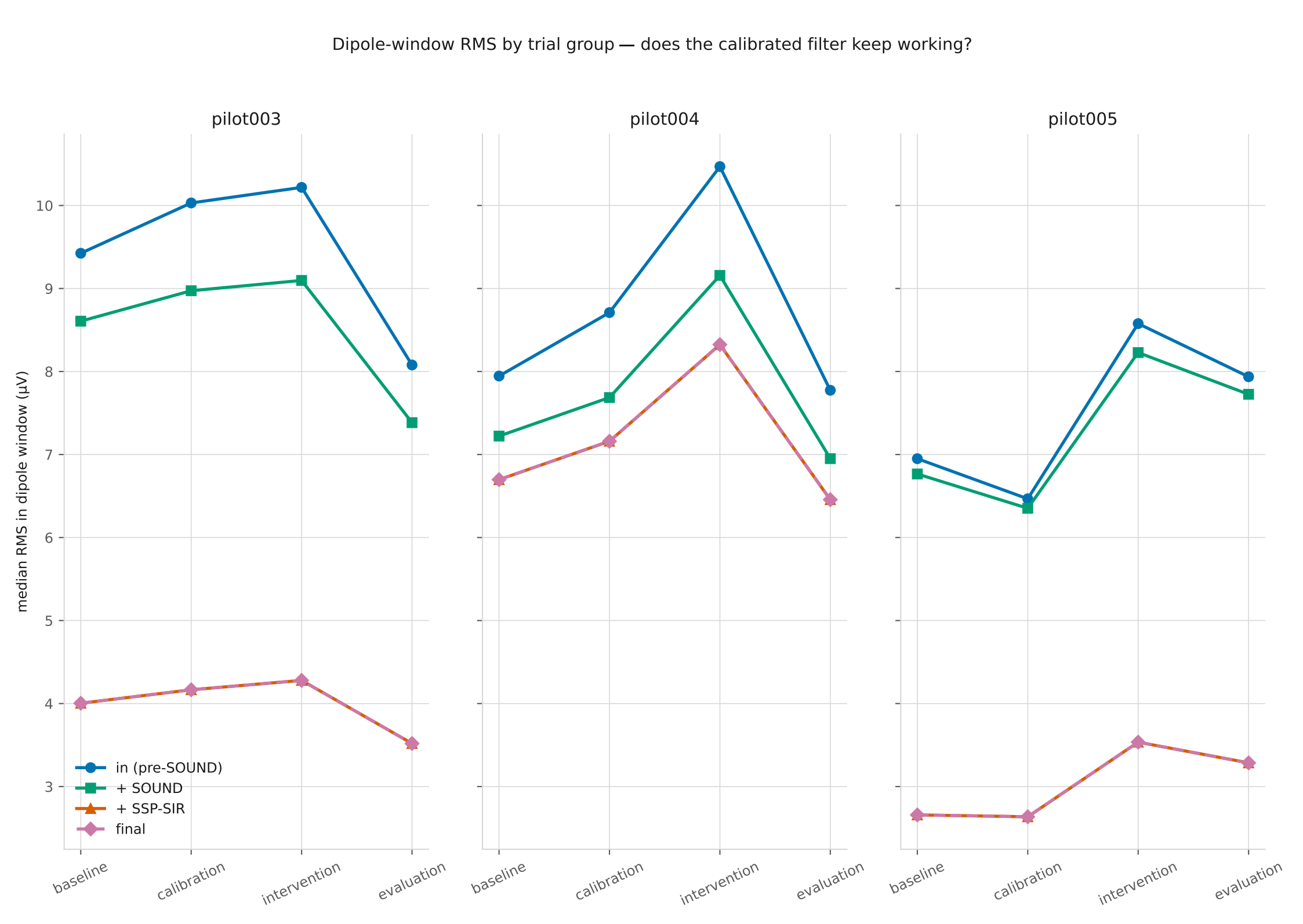

Where each step helps
in (pre-SOUND)
+ SOUND
+ SSP-SIR
final
>100 Hz residual, 1-15 ms
RMS (µV)
0
1
2
3
4
5
pilot003
pilot004
pilot005
evoked / single-trial RMS in dipole window
SNR
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40
pilot003
pilot004
pilot005
effective dimensions, dipole window
participation ratio
0
1
2
3
4
5
6
7
pilot003
pilot004
pilot005


Effect on the fitted dipole (free orientation) and fit quality
pilot003
pilot004
pilot005
amplitude (µA·m)
R²
0.30
0.25
0.20
0.15
0.10
0.05
0.00
1.0
0.8
0.6
0.4
0.2
0.0
in (pre-SOUND)
+ SOUND
+ SSP-SIR
final

Per-trial effect of each step inside the dipole window




Dipole-window RMS by trial group — does the calibrated filter keep working?
pilot003
pilot004
pilot005
median RMS in dipole window (µV)
10
9
8
7
6
5
4
3
in (pre-SOUND)
+ SOUND
+ SSP-SIR
final
baseline
calibration
intervention
evaluation